

Student Number: _____



Roseville College

3 UNIT MATHEMATICS

**Trial Higher School Certificate
1998**

Time Allowed: 2 hours - five minutes for reading

Directions to Candidates

- * Attempt all questions.
- * All necessary working must be shown.
- * Marks may be deducted for careless or badly arranged work.
- * Show your candidate number on each page of your work.
- * Begin each question on a new page.
- * All questions are of equal value.

Question 1

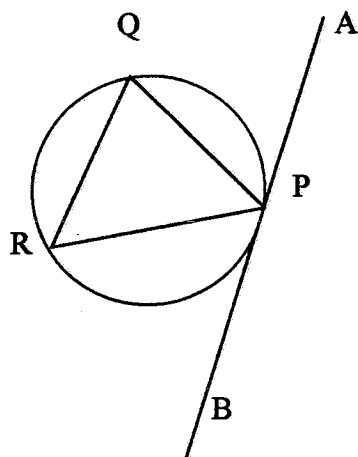
- (a) (i) For what value of k is the polynomial $P(x) = x^3 + 2x^2 - x + k$ divisible by $(x - 2)$? (3)

(ii) Show that $P(x) = 0$ has only one root for that value of k .

- (b) Find the derivative of $y = e^{\sin 3x}$. (2)

- (c) If $f(x) = \frac{1}{x+3}$, find the inverse function, $f^{-1}(x)$. (2)

- (d) (3)



Given that $PQ = PR$ and AB is a tangent to the circle PQR at P , prove that $RQ \parallel BA$.

- (e) Use the substitution $u = 2x^2 - 5$ to find $\int \frac{x}{\sqrt{2x^2 - 5}} dx$ (2)

Question 2 (Start a new page)

- (a) Solve $x - 5 < \frac{14}{x}$ (2)
- (b) Differentiate $y = 5 \tan^{-1} \frac{x}{2}$ (2)
- (c) Find the coefficient of x^3 in the expansion of $(3x + 2)^7$ (2)
- (d) Find the coordinates of the point which divides the interval joining A (3,-2) and B (-1,1) externally in the ratio 3:2. (2)
- (e) Prove $\frac{\sin 2A}{1 - \cos 2A} = \cot A$ and hence obtain an exact value for $\cot 67\frac{1}{2}^\circ$ in simplest surd form. (4)

Question 3 (Start a new page)

- (a) If α, β, δ are the roots of $2x^3 - x^2 - x - 2 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\delta}$. (2)
- (b) Use one application of Newton's method to solve $f(x) = \cos x - \ln x$ given that there is a root near $x = 1$. (2)
- (c) If the volume of a cube is increasing at the rate of $25 \text{ mm}^3/\text{s}$, find the increase in its surface area when its side is 12 mm. (3)
- (d) Sketch the graph of $y = 3 \cos^{-1} 2x$ (at least one third of a page). Indicate the domain and range clearly on your axes. (2)
- (e) The curves $y = \sin x$ and $y = \cos x$ intersect at $x = \frac{\pi}{4}$. If θ is the acute angle between the tangents to the curve $y = \sin x$ and $y = \cos x$, at the point of intersection, find θ (to the nearest degree). (3)

Question 4 (Start a new page)

(a) $\int_2^4 \frac{x}{(3x-4)} dx$ using the substitution $u = 3x - 4$. (2)

(b) Solve the equation $\cos x - \sqrt{3} \sin x = 2$ for $0 \leq x \leq 2\pi$. (3)

(c) The velocity vcm/s of a particle moving in Simple Harmonic Motion along the x axis is given by $v^2 = 72 - 12x - 4x^2$. (4)

(i) Between which two values of x is the particle oscillating?

(ii) What is the amplitude of the motion?

(iii) Find the acceleration of the particle in terms of x .

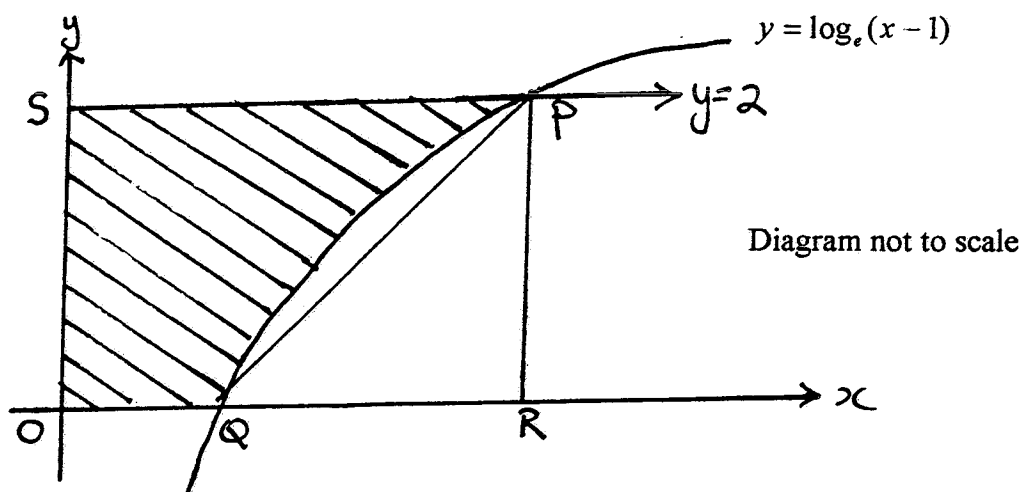
(iv) Find the period of the oscillation.

(d) What is the general solution for $\tan \phi = -\sqrt{3}$? (2)

(e) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{4x}$ (1)

- (b) The graph of the function $y = \log_e(x - 1)$ meets the line $y = 2$ at P and the x axis at Q. From P perpendiculars are drawn to the x axis and y axis meeting them at R and S respectively.

(5)



- (i) Show that the coordinates of P are $(e^2 + 1, 2)$ and write down the coordinates of the points R, S and Q.
- (ii) Show that the shaded area enclosed by the arc PQ, the y axis and the lines $y = 2$ and $y = 0$ divides the rectangle OSPR into 2 portions of equal areas.
- (iii) Show that the area enclosed by the arc QP and the straight line interval QP equals the area of triangle OSQ.

Question 7 (Start a new page)

- (a) Mr Ryan hits a golf ball from a point O with an initial velocity of v m/s so that it rises at an angle of 30° to the horizontal.

(7)

- (i) Show that $x = \frac{\sqrt{3}}{2}vt$, and $y = -5t^2 + \frac{1}{2}vt$, where x and y are the horizontal and vertical displacements of the ball in metres from O, t seconds after the ball has been hit. Take $g = -10\text{m/s}^2$.

The ball lands on a horizontal green 24 metres below O, after a flight of 4 seconds.

- (ii) Show that $v = 28$ m/s
- (iii) Find the greatest height reached by the ball
- (iv) Find the cartesian equation of the trajectory of the ball
- (v) Find the horizontal distance that the ball travelled.
- (b) See over for part (b)

Question 6 (Start a new page)

(a) Given that $(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$ (4)

(i) Show that $\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$

(ii) By differentiating both sides show that $\sum_{k=1}^{2n} k \binom{2n}{k} = n4^n$

(b) Evaluate $\int_0^{\pi/6} \sin^5 x \cos x \, dx$ using the substitution $u = \sin x$ (2)

(c) A meteorite, soon after impact had a temperature of $2,520^\circ\text{C}$, and cooled to $1,950^\circ\text{C}$ in 20 minutes when the surrounding temperature was -20°C . How long would the meteorite take to cool to 0°C ? (Give your answer correct to the nearest minute). (3)

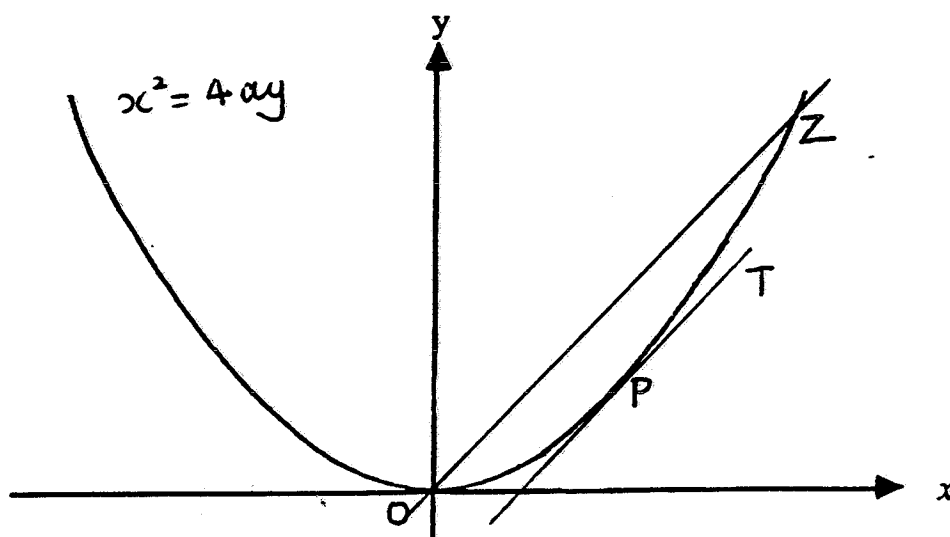
(d) In order to calculate the height of a mountain peak, a surveyor measured the angle of elevation from a certain stake and found it to be $18^\circ 40'$. He then walked 780 m over a level plain towards the mountain and set a second stake from which the angle of elevation was found to be $22^\circ 8'$. Find the height of the peak. (3)

Question 5 (Start a new page)

- (a) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. Tangent PT is drawn at P. (6)

A straight line is drawn, parallel to this tangent and through the origin O. This cuts the parabola again at Z.

- (i) What is the equation of the line OZ?
- (ii) Show that Z is the point $(4ap, 4ap^2)$
- (iii) Find the coordinates of the point, M the midpoint of PZ
- (iv) Find the equation of the locus of M as P moves around the parabola.



- (b) Prove by Mathematical Induction that $8^n - 5^n$ is divisible by 3 for all positive integers n. (4)

- (c) Find the term which is independent of x in the expansion of $\left(2x^3 + \frac{1}{3x^2}\right)^5$. (2)

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1a) i) $P(x) = x^3 + 2x^2 - x + R$

$P(2) = 8 + 8 - 2 + R = 0 \therefore R = -14$

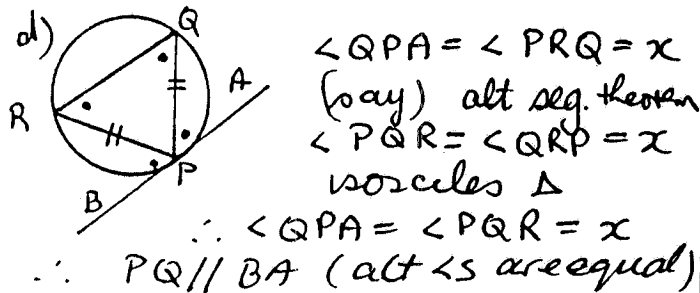
$\begin{array}{r} x^2 + 4x + 7 \\ x-2 \overline{) x^3 + 2x^2 - x - 14} \\ \underline{x^3 - 2x^2} \\ 4x^2 - x \\ \underline{4x^2 - 8x} \\ 7x - 14 \\ \underline{7x - 14} \\ 0 \end{array}$	Δ of $x^2 + 4x + 7$ $= 16 - 28$ $= -ve$ $\therefore$ no real root
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$\therefore P(x)$ has only 1 real root.

b) $y = e^{\sin 3x} \quad y' = 3 \cos 3x e^{\sin 3x}$

c) $y = \frac{1}{x+3} \quad f^{-1}(x) \quad x = \frac{1}{y+3}$

$\frac{1}{x} = y+3 \therefore y = \frac{1}{x} - 3$



e) $\int \frac{x}{\sqrt{2x^2-5}} dx$

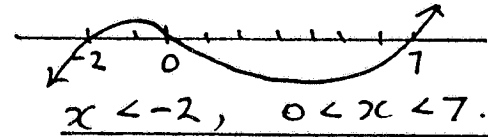
$u = 2x^2 - 5$
 $\frac{du}{dx} = 4x$
 $= \frac{1}{4} \int u^{-1/2} du = \frac{1}{4} 2u^{1/2} + C$
 $= \frac{1}{2} \sqrt{2x^2-5} + C$

2a) $x - 5 < \frac{14}{x}, \quad x \neq 0$

$x^2(x-5) < 14x$

$x^3 - 5x^2 - 14x < 0$

$x(x-7)(x+2) < 0$



b) $y = 5 \tan^{-1} \frac{x}{2} \quad y' = \frac{10}{4+x^2}$

c) $(3x+2)^7 T_5 = {}^7C_4 (3x)^3 2^4$
 coefficient = 15120

d) $A(3, -2) \quad B(-1, 1) \quad 3:2$

$x = \frac{mx_2 + nx_1}{3-2} = \frac{3(-1) + 2(-2)}{1} = -9$

$y = \frac{my_2 + ny_1}{3-2} = \frac{3(1) + 2(-2)}{1} = 7$
Point = (-9, 7)

e) Prove $\frac{\sin 2A}{1 - \cos 2A} = \cot A$

L.H.S. $= \frac{2 \sin A \cos A}{1 - (\cos^2 A - \sin^2 A)} = \frac{2 \sin A \cos A}{1 - \cos^2 A + \sin^2 A}$

$= \frac{2 \sin A \cos A}{2 \sin^2 A} = \frac{\cos A}{\sin A} = \cot A = R.H.S.$

$\cot 67\frac{1}{2}^\circ = \frac{\sin 135^\circ}{1 - \cos 135^\circ} = \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$

$= \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{2}+1}{\sqrt{2}}} = \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = \frac{\sqrt{2}-1}{1} = \sqrt{2}-1$

$\frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1} = \frac{\sqrt{2}-1}{1} = \sqrt{2}-1$

3a) $2x^3 - x^2 - x - 2 = 0$

$\frac{1}{2} + \frac{1}{3} + \frac{1}{8} = \frac{38 + 27 + 12}{216}$

$= -\frac{1}{2} \div 1 = -\frac{1}{2}$

b) $a_2 = a_1 - \frac{f(a_1)}{f'(a_1)}$

$f(x) = \cos x - \ln x$

$f'(x) = -\sin x - \frac{1}{x}$

$a_2 = 1 - \frac{\cos 1 - 0}{-\sin 1 - 1}$

$= 1 - \frac{0.5403023}{-1.841471}$

≈ 1.293408

root ≈ 1.29 (2 d.p.)

c) $V = s^3 \quad \frac{dV}{ds} = 3s^2 \quad \frac{dV}{ds} \times \frac{ds}{dt} = 25 \text{ mm}^3/\text{s}$

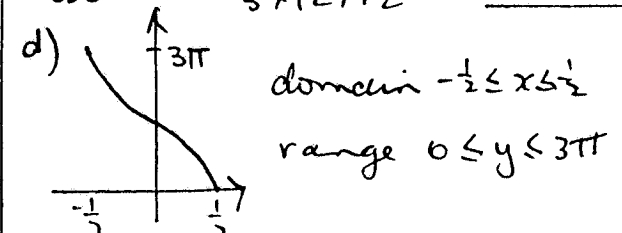
$A = 6s^2 \quad \frac{dA}{ds} = 12s \quad \frac{dA}{ds} \times \frac{ds}{dt}$

$3s^2 \times \frac{ds}{dt} = 25 \therefore \frac{ds}{dt} = 25 \div 3s^2$

$\frac{dA}{dt} = \frac{dA}{ds} \cdot \frac{ds}{dt} = 12s \times 25 \div 3s^2$

when $s = 12$.

$\frac{dA}{dt} = \frac{12 \times 12 \times 25}{3 \times 12 \times 12} = 8\frac{1}{3} \text{ mm}^2/\text{s}$



$$3e \quad y = \sin x \quad y' = \cos x$$

$$y = \cos x \quad y' = -\sin x$$

$$\tan \theta = \frac{\frac{1}{\sqrt{2}} - -\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}}}$$

$$= \frac{1.41421356}{1 - 0.5}, \theta \doteq 71^\circ$$

$$4a) \int_2^4 \frac{x \, dx}{3x-4} \quad x=2, u=2$$

$$x=4, u=8$$

$$= \frac{1}{3} \cdot \frac{1}{3} \int_2^8 \frac{u+4}{u} \, du \quad u=3x-4$$

$$= \frac{1}{9} [u + 4 \ln u]_2^8 \quad \frac{du}{dx} = 3$$

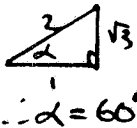
$$= \frac{1}{9} \left(\{8 + 4 \ln 8\} - \{2 + 4 \ln 2\} \right) \quad \frac{u+4}{3} = x$$

$$= \frac{1}{9} \left\{ \frac{1}{6} + 4 \ln 4 \right\} \doteq 1.28 \text{ (2 d.p.)}$$

$$b) \cos x - \sqrt{3} \sin x = 2$$

$$\cos(x+\alpha) = 1$$

$$\cos(x+\frac{\pi}{3}) = 2\pi$$

$$x = \frac{5\pi}{3}$$


$$\therefore \alpha = 60^\circ$$

$$c) i) v^2 = 72 - 12x - 4x^2$$

$$0 = 4(18 - 3x - x^2) = 4(6+x)(3-x)$$

$$\therefore x = -6 \text{ and } +3$$

$$ii) \text{amplitude} = \frac{3-(-6)}{2} = \frac{4\frac{1}{2}}{2}$$

$$iii) \text{accel} = \frac{d^2v^2}{dx^2} = \frac{d}{dx}(36 - 6x - 2x^2)$$

$$= -6 - 4x$$

$$iv) -6 - 4x = -4(\frac{3}{4} + x) \therefore n = \sqrt{4} = 2$$

$$\text{period} = \frac{2\pi}{2} = \pi \text{ secs}$$

$$d) \tan \phi = -\sqrt{3} \therefore \phi = -\pi/3$$

$$\therefore \phi = n\pi - \pi/3$$

$$e) \lim_{x \rightarrow 0} \frac{\sin 2x}{4x} = \lim_{x \rightarrow 0} \frac{1}{2} \frac{\sin 2x}{2x}$$

$$= \frac{1}{2} \times 1 = \frac{1}{2}$$

$$5a) i) P(2ap, ap^2) \quad x^2 = 4ay$$

$$\frac{dy}{dx} = \frac{2x}{4a} \quad m = \frac{2 \cdot 2ap}{4a} = p$$

$$\therefore \text{gradient of tangent} = p$$

$$02 = y - 0 = p(x - 0) \therefore y = px$$

$$ii) x^2 = 4ay \quad y = px$$

$$x^2 = 4apx \quad 0 = x^2 - 4apx$$

$$x(x - 4ap) \therefore x = 0 \text{ or } 4ap$$

$$px = y \therefore x = y/p \quad 4ay = y^2/p^2$$

$$4ap^2y = y^2 \therefore y^2 - 4ap^2y = 0$$

$$y(y - 4ap^2) = 0 \therefore y = 0 \text{ or } 4ap^2$$

$$\therefore 2 = (4ap, 4ap^2)$$

$$iii) M = \left(\frac{2ap + 4ap}{2}, \frac{ap^2 + 4ap^2}{2} \right)$$

$$= (3ap, \frac{5}{2}ap^2)$$

$$iv) x = 3ap \therefore p = x/3a$$

$$y = \frac{5}{2}ap^2 \therefore y = \frac{5a}{2} \cdot \frac{x^2}{9a^2}$$

$$y = \frac{5x^2}{18a} \text{ or } 18ay = 5x^2$$

5b) Prove $8^n - 5^n$ is divisible by 3 for all positive n .

① Test for $n=1$, $8^1 - 5^1 = 3$ which is divisible by 3

② Assume it is true for $n=k$ i.e. $8^k - 5^k = 3P$ where P is an integer

③ Test for $n=k+1$
 $8^{k+1} - 5^{k+1} = 8 \cdot 8^k - 5 \cdot 5^k$
 $= 8 \cdot 8^k - 8 \cdot 5^k + 3 \cdot 5^k = 8(8^k - 5^k) + 3 \cdot 5^k$
 $= 8 \cdot 3P + 3 \cdot 5^k = 3(8P + 5^k)$ which is divisible by 3 $\therefore$ true for $n=k+1$ if true for $n=k$

④ Since it is true for $n=1$, then it is true for $n=1+1=2$, $2+1=3$ etc for all positive n .

$$c) (2x^3 + \frac{1}{3x^2})^5 \quad T_{n+1} = {}^nC_n (2x^3)^{5-n} (\frac{1}{3x^2})^n$$

$$(x^3)^{5-n} \cdot x^{-2n} = x^0$$

$$15 - 3n + -2n = 0 \quad 15 - 5n = 0 \therefore n=3$$

$$T_4 = {}^5C_3 (2x^3)^2 (\frac{1}{3x^2})^3 = \frac{10 \times 4}{27} = \frac{40}{27}$$

$$6a) (1+x)^{2n} = \sum_{k=0}^{2n} {}^{2n}C_k x^k$$

i) Show that $\sum_{k=0}^{2n} {}^{2n}C_k = 4^n$

$$\text{Let } x=1$$

$$\text{L.H.S.} = (1+1)^{2n} = 2^{2n} = 4^n$$

$$\text{RHS} = \sum_{k=0}^{2n} {}^{2n}C_k 1^k = \sum_{k=0}^{2n} {}^{2n}C_k$$

$$\therefore \sum_{k=0}^{2n} {}^{2n}C_k = 4^n$$

6a) ii) Differentiating

L.H.S. $2n(1+x)^{2n-1}$

$$\text{RHS} = \binom{2n}{1} + 2\binom{2n}{2}x + 2\binom{2n}{3}x^2 + \dots + 2n\binom{2n}{2n}x^{2n-1}$$

Let $x=1$

$$\text{LHS} = 2n(1+1)^{2n-1} = 2n(2)^{2n-1} = n(2n)^{2n} = n4^n$$

$$\text{RHS} = \binom{2n}{1} + 2\binom{2n}{2} + 3\binom{2n}{3} + \dots + 2n\binom{2n}{2n}$$

$$= \sum_{k=1}^{2n} k \binom{2n}{k}$$

$$\therefore \sum_{k=1}^{2n} k \binom{2n}{k} = n4^n$$

$$b) \int_0^{\pi/6} \sin^5 x \cos x dx$$

$$u = \sin x \\ \frac{du}{dx} = \cos x$$

$$x=0, u=\sin 0=0 \\ x=\pi/6, u=\sin \pi/6 = \frac{1}{2}$$

$$\int_0^{1/2} u^5 du \\ = \left[\frac{u^6}{6} \right]_0^{1/2} = \frac{1}{384}$$

$$c) T = -20 + Ae^{-kt}$$

$$2520 = -20 + Ae^0$$

$$2540 = A$$

$$T = -20 + 2540e^{-kt}$$

$$1950 = -20 + 2540e^{-20k}$$

$$1970 = 2540e^{-20k}$$

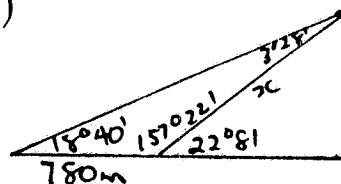
$$\frac{1970}{2540} = e^{-20k}$$

$$\therefore k = 0.012706526$$

$$\ln \frac{20}{2540} = -kt$$

$$\therefore t = 381 \text{ minutes}$$

6d)

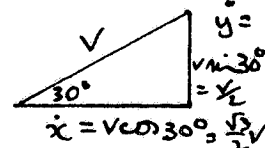
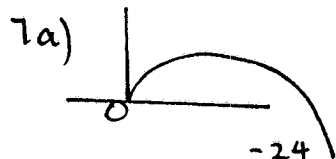


$$\frac{x}{\sin 18^\circ 40'} = \frac{780}{\sin 30^\circ 28'}$$

$$x = \frac{780 \times \sin 18^\circ 40'}{\sin 30^\circ 28'} = 4128.6 \text{ m}$$

$$\sin 22^\circ 8' = \frac{h}{4128.6}$$

$$h = \sin 22^\circ 8' \times 4128.6 = 1555.5 \text{ m}$$



$$i) \dot{x} = 0$$

$$\dot{x} = \int 0 dt = 0 + C$$

$$\text{at } t=0, \dot{x} = \frac{\sqrt{3}}{2} v$$

$$\therefore \dot{x} = \frac{\sqrt{3}}{2} v$$

$$x = \int \frac{\sqrt{3}}{2} v dt = \frac{\sqrt{3}}{2} vt + C_1$$

$$\text{at } t=0, x=0 \therefore C_1=0$$

$$x = \frac{\sqrt{3}}{2} vt$$

$$\ddot{y} = -10 \therefore \dot{y} = \int -10 dt = -10t + C_2$$

$$\text{at } t=0, \dot{y} = \frac{v}{2} \therefore \dot{y} = -10t + \frac{v}{2}$$

$$y = \int -10t + \frac{v}{2} dt = -5t^2 + \frac{v}{2}t + C_3$$

$$\text{at } t=0, y=0 \therefore y = -5t^2 + \frac{v}{2}t$$

$$ii) y = -5t^2 + \frac{v}{2}t \text{ when } t=4, y=-24$$

$$-24 = -5(16) + \frac{4v}{2}$$

$$-24 = -80 + 2v$$

$$56 = 2v \therefore v = 28 \text{ m/s}$$

iii) greatest height occurs when

$$\dot{y} = 0$$

$$\dot{y} = -10t + \frac{v}{2} = 0$$

$$-10t + \frac{28}{2} = 0$$

$$\therefore -10t = -14 \therefore t = 1.4 \text{ sec}$$

$$y = -5t^2 + 14t$$

$$\text{when } t=1.4, y = -5(1.4)^2 + 14(1.4) = 9.8 \text{ m}$$

$$iv) x = \frac{\sqrt{3}}{2} \cdot 28t \quad y = -5t^2 + 14t$$

$$\text{from ① } t = \frac{x}{14\sqrt{3}} \text{ sub into ②}$$

$$y = -5\left(\frac{x}{14\sqrt{3}}\right)^2 + 14\left(\frac{x}{14\sqrt{3}}\right) \\ = -\frac{5x^2}{588} + \frac{\sqrt{3}x}{3}$$

$$v) x = \frac{\sqrt{3}}{2} vt = \frac{\sqrt{3}}{2} \cdot 28 \cdot 4 \\ = 97 \text{ m}$$

$$7b) y = \log_e(x-1)$$

$$2 = \log_e(x-1)$$

$$e^2 = x-1 \quad \therefore x = e+1 \therefore P(e+1, 2)$$

$$R = (e^2+1, 0), \quad S(0, 2) \quad Q(2, 0)$$

$$ii) y = \log_e(x-1)$$

$$e^y = x-1 \quad \therefore x = e^y+1$$

$$\int_0^2 (e^y+1) dy = [e^y+y]_0^2$$

$$= (e^2+2) - (e^0+0) = e^2+2-1 = (e^2+1)u^2$$

$$\text{Area rectangle OSPR} = 2 \times (e^2+1)u^2$$

$$\therefore \text{shaded area} = \frac{1}{2} \text{ rectangle}$$

$$iii) \Delta QRP = \frac{1}{2} \times 2 (e^2+1-2) = e^2-1u^2$$

$$\begin{aligned} \text{Area of segment} &= (e^2+1) - (e^2-1) \\ &= 2u^2 \end{aligned}$$

$$\text{Area of } \Delta OSQ = \frac{1}{2} \times 2 \times 2 = 2u^2$$

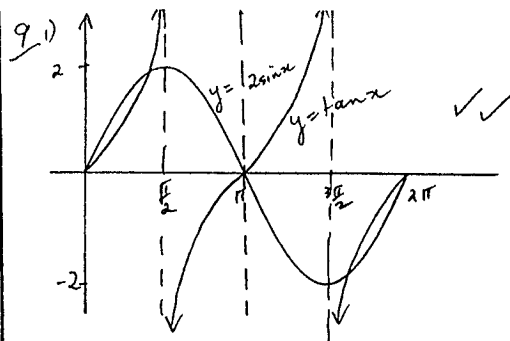
$$\therefore \text{Area of segment} = \Delta OSQ = 2u^2$$

a) i) Area $\div \frac{2}{3} \left\{ 1 + 4 \times \frac{\sqrt{3}}{2} + 0 \right\}$
 $= \frac{2}{3} (1 + 2\sqrt{3})$ units²
 $\div 2.976$ units²

ii) $V = \pi \int_{-2}^2 \left(\frac{1}{2} \sqrt{4-x^2} \right)^2 dx$
 $= 2\pi \int_0^2 \frac{4-x^2}{4} dx$
 $= \frac{\pi}{2} \int_0^2 (4-x^2) dx$
 $= \frac{\pi}{2} \left[4x - \frac{x^3}{3} \right]_0^2$
 $= \frac{\pi}{2} \left(8 - \frac{8}{3} - (0-0) \right)$
 $= \frac{8\pi}{3}$ units³

b) $A = 80e^{-0.025t}$
 i) $t=10$, $A = 80e^{-0.25}$
 $\div 62.3g$
 ii) Initial amt = 80g
 Half life $\rightarrow 40g$
 $40 = 80e^{-0.025t}$
 $\frac{1}{2} = e^{-0.025t}$
 $\ln \frac{1}{2} = -0.025t$
 $t = \frac{\ln \frac{1}{2}}{-0.025}$
 $\div 27.73$ years

iii) $\frac{dA}{dt} = 80 \times -0.025 e^{-0.025t}$
 $= 2e^{-0.025t}$
 $t=8$ $\frac{dA}{dt} = 2e^{-0.2} \div 1.649/\text{year}$



ii) There are 5 solutions to $2\sin x = \tan x$ from the graph

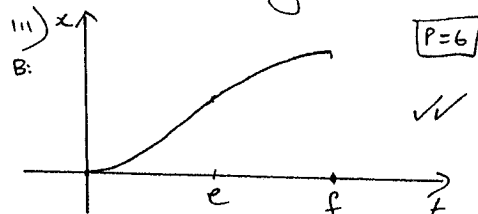
iii) $2\sin x = \tan x$
 $2\sin x = \frac{\sin x}{\cos x}$
 $2\sin x \cos x = \sin x$
 $2\sin x \cos x - \sin x = 0$
 $\sin x (2\cos x - 1) = 0$

iv) $\sin x = 0$, $2\cos x - 1 = 0$
 $x = 0, \pi, 2\pi$ $\cos x = \frac{1}{2}$
 $x = \frac{\pi}{3}, \frac{5\pi}{3}$

$\therefore$ Solns are $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$

b) i) A shows the greater accⁿ since gradients of tangents for A are always greater than gradients of tangents for B & $\frac{dy}{dt}$ = acceleration

ii) Both A & B have the same velocity at $t=e$, however B is accelerating and A is decelerating.



10 a) i) $OR = x$
 $\therefore QR = e^{-x^2}$
 $\therefore \text{Area PQRS} = 2x \times e^{-x^2}$
 $= 2xe^{-x^2}$

ii) $\frac{dA}{dx} = 2x \times -2xe^{-x^2} + 2e^{-x^2}$
 $= -4x^2e^{-x^2} + 2e^{-x^2}$
 $= 2e^{-x^2} (1 - 2x^2)$

For max Area, $\frac{dA}{dx} = 0$

$\therefore 2e^{-x^2} (1 - 2x^2) = 0$

$2e^{-x^2} \neq 0$, $1 - 2x^2 = 0$
 $2x^2 = 1$
 $x^2 = \frac{1}{2}$
 $x = \pm \frac{1}{\sqrt{2}}$

But initially $x > 0 \therefore x = \frac{1}{\sqrt{2}}$

Check max

$\frac{d^2A}{dx^2} = 2e^{-x^2} (-4x) + 4xe^{-x^2} (1 - 2x^2)$
 $= -8xe^{-x^2} - 4xe^{-x^2} + 8x^3e^{-x^2}$
 $= 4xe^{-x^2} (2x^2 - 3)$

When $x = \frac{1}{\sqrt{2}}$, $2x^2 - 3 = 2 \times \frac{1}{2} - 3 = -1 < 0$
 $\therefore \text{max} (4xe^{-x^2} > 0)$

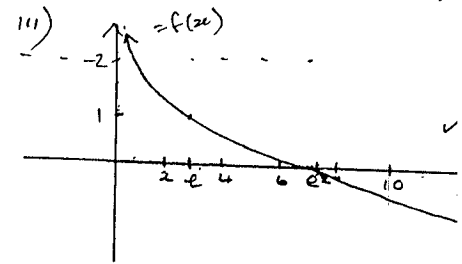
$\therefore x = \frac{1}{\sqrt{2}}$ gives max area of PQRS

b) i) $f(x) = 2 - \log_e x$
 $f'(x) = -\frac{1}{x} < 0$ for all x

($\log_e x$ only exists for $x > 0$)
 $\therefore f(x)$ is always decreasing

ii) x intercept, $f(x) = 0$
 ie $0 = 2 - \ln x$
 $\ln x = 2$
 $x = e^2$

$\therefore x$ intercept = $(e^2, 0)$



iv) $A = \int_0^2 f(y) dy$

$y = 2 - \ln x$
 $\ln x = 2 - y$
 $x = e^{2-y}$

$\therefore A = \int_0^2 e^{2-y} dy$